

Software Solution Requirements Gap Analysis

Librika (e-Pathshala) Platform vs. Digital Library & LMS Specifications

Section	Total Requirements	Fully Met	Partially Met	Not Met
1. Smart Library Solution	9	5	2	2
2. ILMS & LMS Modules	15	8	2	5
3. Mobile Access Platform	5	3	0	2
4. Analytics & Dashboard	7	5	1	1
5. AI-Enabled Features	6	2	0	4
Overall Summary	42	23 (55%)	5 (12%)	14 (33%)

1. Smart Library Solution

RFID-Based Library Management & Asset Tracking System

- **RFID-enabled library management:** **Not Met.** The system operates without RFID reader hardware integrations.
- **Asset tracking system:** **Partially Met.** Tracks physical checkout transactions and barcode identities, but lacks real-time localization tags.
- **Membership management:** **Fully Met.** Supports user profiles, registration, and permissions.
- **Book issue/return/renewal management:** **Fully Met.** Comprehensive circulation records, checkout terms, renewals, and overdue logs.
- **Cataloguing and accession management:** **Fully Met.** Captures title, author, publisher, subject, category, and standard barcode tags.
- **Inventory management:** **Fully Met.** Auto-tracks available copies and physical stock count changes.
- **OPAC (Online Public Access Catalogue):** **Fully Met.** Students browse books and search categories online.
- **Reporting and analytics:** **Fully Met.** Dashboard tracks active checkouts, overdues, and catalog stock totals.
- **Barcode and RFID integration:** **Partially Met.** Generates Code-128 barcode labels for scanning, but lacks RFID support.

2. Integrated Library (ILMS) & Learning Management (LMS)

Library Management Module

- **Acquisition management:** **Fully Met.** Add physical inventory via forms or smart AI cover vision recognition.
- **Cataloguing:** **Fully Met.**
- **Circulation management:** **Fully Met.**
- **User management:** **Fully Met.**
- **Digital repository:** **Fully Met.** Hosts and displays student notes, projects, and PDFs.
- **OPAC search:** **Fully Met.**
- **Reports and analytics:** **Fully Met.**

Learning Management Module (LMS)

- **User registration and authentication:** **Fully Met.**
- **Course management:** **Not Met.** No course curricula or lesson modules.
- **Video learning platform:** **Not Met.** System is reading-centric (PDF/EPUB) and lacks video player tools.
- **Assessment and testing modules:** **Partially Met.** Book quiz generation tests reading comprehension, but general course tests are absent.
- **Certification management:** **Not Met.** No completion certificates.
- **Progress tracking:** **Partially Met.** Tracks e-reading book percentage and streaks, not courses.
- **Discussion forums:** **Not Met.** No forums, although user book reviews are supported.
- **Role-based access management:** **Fully Met.**

3. Mobile Access Platform

- **Android application:** **Not Met.** Web application without native packaging.
- **iOS application:** **Not Met.** Web application without iOS packaging.
- **Responsive web portal:** **Fully Met.** Mobile-friendly design featuring a native-feeling mobile bottom nav bar.
- **Mobile-friendly access:** **Fully Met.** Visual interfaces scale automatically to phones and tablets.
- **User dashboard and notifications:** **Fully Met.**

4. Analytics and Monitoring Dashboard

- **Usage monitoring:** **Fully Met.**
- **User engagement analytics:** **Fully Met.**
- **Learning progress tracking:** **Partially Met.** Logs book progress percentage.
- **Administrative dashboards:** **Fully Met.**
- **Performance indicators:** **Fully Met.**
- **MIS and reporting tools:** **Fully Met.**
- **Exportable reports:** **Not Met.** Admin analytics displayed on-screen, but no CSV/PDF export exists.

5. AI-Enabled Features

- **AI chatbot/virtual assistant:** **Fully Met.** Active chatbot functionality integrated via /api/chat.
- **AI-powered search:** **Fully Met.** Implements conceptual semantic search matching using NVIDIA completions API.
- **Personalized content recommendations:** **Not Met.** Recommendations filter by genre categories, not AI personalization.
- **Adaptive learning pathways:** **Not Met.**
- **Career guidance and counselling tools:** **Not Met.**
- **AI-based analytics and report:** **Not Met.**